

Basic IPSEC Traffic Trouble shooting on TMS Network .

The TMS IPSEC Tunnel is a has two default setting that enable the Tunnel to remain active while traffic present and go idle when traffic is non-existent. *The IPSEC IpDataOffer RefreshLifetimeProposed (60 min) RefreshLifetimeAccepted (60 min 480Min)* parameters dictate how long a session can remain idle before the tunnel is taken down and refreshed. If during that time an outage or problem occurs on either end where the tunnel is refreshed; the session is flagged as invalid. Typically the resolution is to restart both tunnels simultaneously to synchronize the session. This feature is one of the salient security settings that make IPSec/VPN's so attractive and less vulnerable to attack.

If there are network errors of an undetermined nature and the system is unable to respond, then

- 1) A restart of the PAGENT, TRD Daemon (nee a Started Task) , and Policy Agents is performed
- 2) The end user on the other endpoint is asked to respond in a similar fashion
- 3) TCPIP on the mainframe is required

A typical traffic scenario as captured in the TCPIP SYSTEM LOG shows the following routine incidents. Much the detail comes from the TRMD which generates information on tunnel activity and traffic. This information while useful does not easily lend itself to trouble shooting traffic problems. As long as the tunnel is active this informs the reader were the traffic is initiating and terminating on the tunnel.

Mar 31 19:54:56 TMSN1 TRMD.TCPIP[17]: EZD0814I Packet permitted: 03/31/2009 19:54:48.13 filter rule= Sterling1~7 ext= 2 sipaddr= 209.95.224.31 dipaddr= 150.45.37.1 proto= esp(50) -= -= -= Interface= 10.52.9.10 (I) secclass= 255 dest= local len= 152 vpnaction= N/A tunnelID= N/A

Mar 31 19:54:56 TMSN1 TRMD.TCPIP[17]: EZD0814I Packet permitted: 03/31/2009 19:54:48.13 filter rule= Sterling2~9 ext= 2 sipaddr= 209.95.252.101 dipaddr= 150.45.37.1 proto= tcp(6) sport= 1334 dport= 35556 -= Interface= 10.52.9.10 (I) secclass= 255 dest= local len= 101 vpnaction= TMS_IPSec_Sterling tunnelID= Y9

Mar 31 19:54:56 TMSN1 TRMD.TCPIP[17]: EZD0814I Packet permitted: 03/31/2009 19:54:48.13 filter rule= Sterling2~9 ext= 1 sipaddr= 150.45.37.1 dipaddr= 209.95.252.101 proto= tcp(6) sport= 35556 dport= 1334 -= Interface= 10.52.9.2 (O) secclass= 255 dest= local len= 84 vpnaction= TMS_IPSec_Sterling tunnelID= Y9

Mar 31 19:54:56 TMSN1 TRMD.TCPIP[17]: EZD0814I Packet permitted: 03/31/2009 19:54:48.14 filter rule= Sterling1~7 ext= 2 sipaddr= 209.95.224.31 dipaddr= 150.45.37.1 proto= esp(50) -= -= -= Interface= 10.52.9.10 (I) secclass= 255 dest= local len= 104 vpnaction= N/A tunnelID= N/A

Likewise if the user is able to view IP traffic in the OMVS STDERR data set but care must be taken as the STDERR log file is also very large. Once the log is located the viewer can browse the contents.

Aug 17 03:43:36 TMSN1 IKE: Message instance 327: Phase 1 SAID : 7

Aug 17 03:43:36 TMSN1 IKE: Message instance 327: Stackname : TCPIP

Aug 17 03:43:36 TMSN1 IKE: Message instance 327: Local IKE ID info : Ipv4 69.25.174.250

Aug 17 03:43:36 TMSN1 IKE: Message instance 327: Remote IKE ID info : Ipv4 207.166.201.126

Aug 17 03:43:36 TMSN1 IKE: Message instance 327: Local IKE IP : 150.45.37.1 Port : 500

Aug 17 03:43:36 TMSN1 IKE: Message instance 327: Remote IKE IP : 207.166.201.126 Port : 500

Aug 17 03:43:36 TMSN1 IKE: Message instance 327: KeyExchangeRuleName : ecediltd~5

Aug 17 03:43:36 TMSN1 IKE: Message instance 327: Icookie/Rcookie : xFDD03061C6DFC87E / x3031987AB3BC910D

Aug 17 03:43:36 TMSN1 IKE: Message instance 327: EZD0984I IKE function 0AC5 - pre_process_phaseII_msg() failed : 0 | 0 |

Aug 17 03:43:46 TMSN1 IKE: Message instance 328: EZD0953I Filter installation failed due to conflict with existing filter

Aug 17 03:43:46 TMSN1 IKE: Message instance 329: SA Information

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There is enough useful information available from the both log files to aid in problem determination, but not to generate alerts. To be useful the reader must be familiar with the installation settings and have some prior knowledge of the recent behavior of the specific tunnel in question. This makes troubleshooting an IPSEC Tunnel a rather unique opportunity, and carries with it a host of pitfalls.

Noting the Mainframe has active tools (NMI) which will generate necessary detail to alert the user but requires heavy expertise in coding programs that examine system level activities. This would go far beyond the scope of any normal departmental second third level specialist. An outside consultant or a Specialist firm would probably be best suited to handle these sort of requirements. While a worthwhile objective it is fairly narrow in scope and would produce a singularly useful piece of software with a specific niche.

The issue that needs to be addressed is the inability to view network traffic at the various stages where the present can readily take action. While there are a plethora of tools and products it would a certain fiasco if TMS had different suits monitor the various version of IP protocols traffic.